

Shrinkage invalidates the Hosmer–Lemeshow test: goodness of fit for penalized logistic regression, with an application to glaucoma diagnosis

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Abstract

Clinical prediction models are increasingly fitted by penalized logistic regression, because collinearity or many candidate predictors makes maximum likelihood unstable or impossible. Calibration is then almost always assessed by a grouped goodness-of-fit test such as the Hosmer–Lemeshow test. We show that this combination is invalid. Under ridge regression the grouped standardized residuals acquire a non-centrality induced by shrinkage, so the reference distribution used in practice is wrong, and at the penalty that most improves the fitted probabilities the test rejects correctly specified models between 92 and 100 per cent of the time. We derive the corrected law, show that subtracting an estimate of the shrinkage non-centrality restores the maximum likelihood reference exactly to first order, and obtain a valid procedure by pre pivoting, at a power cost we measure. We also give the attenuation law governing what any such test can detect once the linear predictor must be estimated. In data on glaucoma diagnosis by confocal laser tomography, where the maximum likelihood estimate does not exist, the corrected test finds the evidence for misfit weaker by more than three orders of magnitude: the fitted risks are too flat rather than mis-ordered, so the model needs recalibration rather than rebuilding.

Keywords calibration; clinical prediction model; glaucoma; goodness-of-fit; Hosmer–Lemeshow test; ridge regression

1. A glaucoma model that cannot be fitted by maximum likelihood

Glaucoma is a leading cause of *irreversible* blindness worldwide, and it is insidious: the disease usually progresses without symptoms until a substantial part of the visual field has already been lost, and what has been lost cannot be recovered (Coan et al., 2023). Everything therefore depends on detecting the disease early, while treatment can still preserve the sight that remains. Diagnosis rests largely on the appearance of the optic nerve head, and expert assessment of it is subjective, slow and expensive (Coan et al., 2023), which is why quantitative imaging and statistical classification have been pursued for three decades.

We consider a diagnostic model of exactly this kind, built on confocal laser scanning tomography of the optic nerve head. The data comprise $n = 196$ eyes, 98 with glaucoma and 98 normal, each summarized by $p = 62$ morphometric variables: areas, volumes, depths and slopes of the optic disc, measured globally and within six sectors, together with height variation and cup shape measures. The clinical question is the ordinary one. Given these measurements for a new eye, what is the probability that it is glaucomatous, and can that probability be trusted enough to act on?

Anyone who develops clinical prediction models will recognize the sequence that follows, because it is the one the reporting guidelines (Collins et al., 2015) and the sample-size literature (Riley et al., 2021) prescribe: penalize when

the number of candidate predictors is large relative to the number of events, tune the penalty by cross-validation, then report calibration, almost always with a calibration plot and a Hosmer–Lemeshow test. This paper is about that sequence. Its last step is not valid after its second, so a model reported as badly calibrated in a development paper may have been condemned for using the estimator that the same guidance recommended.

1.1. Maximum likelihood is not available

The variables are repeated geometric summaries of the same optic disc, so they are collinear by construction rather than by accident: the largest pairwise correlation is 0.996 and the condition number of the cross-product matrix of the standardized predictors is 5.0×10^7 . With $p/n = 0.32$ and collinearity of this order, the logistic likelihood is maximized on a boundary; the iteratively reweighted least squares algorithm does not converge, and the maximum likelihood estimate does not exist.

This is worth stating plainly because it removes a choice that is usually available. The analyst cannot fit this model by maximum likelihood and then decide whether penalization would be an improvement. Penalization is the only route to a fitted model at all. Ridge estimation for logistic regression, in the form we use, is due to le Cessie and van Houwelingen (1992); with the penalty chosen by ten-fold cross-validation it gives a model that discriminates well, with an area under the receiver operating characteristic curve of 0.905, and which a clinician might reasonably consider using.

1.2. The routine check, and what it reports

Having fitted the model, the analyst asks whether it fits. In clinical prediction the near-universal answer is a grouped goodness-of-fit test, and in practice almost always the Hosmer–Lemeshow test (Hosmer and Lemeshow, 1980; Copas, 1989), whose standing rests largely on the comparison study of Hosmer et al. (1997): sort the patients by predicted risk, cut them into ten groups, and compare the number of events observed in each group with the number the model expected. For these data that test returns $p < 10^{-5}$ at the penalty cross-validation selects, and it rejects at the 5% level in every one of the twelve penalty-and-grouping configurations we examined, when the grouped statistic is referred to the maximum likelihood covariance — the sharper of the two forms in use, and the one we compare against throughout (Section 6.2). The textbook χ^2_{G-2} form rejects in eleven of the twelve.

Read in the ordinary way, that is a verdict: the model is badly misspecified, its predicted probabilities cannot be believed, and it should be rebuilt before anyone relies on it.

We show in this paper that the verdict is not available, because the test that produced it is not valid for the model that was fitted. Its reference distribution is derived under maximum likelihood estimation — the very thing these data do not admit — and under penalized estimation it is wrong in a specific and quantifiable way. At the penalty that cross-validation selects, that test rejects models we know to be correct with probability approaching one.

1.3. Why this matters beyond one dataset

Penalized regression is recommended whenever the number of candidate predictors is large relative to the number of events (Riley et al., 2021; Pavlou et al., 2024), and the resulting model is then assessed for calibration in the usual way. Calibration — the agreement between predicted probabilities and observed event rates — deserves that attention, because a model that discriminates well but is miscalibrated can still be misleading, and can be actively harmful when used to guide decisions (Van Calster et al., 2019). Miscalibration always reduces the net benefit of using a model, and for some risk thresholds it makes a model worse than treating everyone or no one (Van Calster and Vickers, 2015). Across 158 external validations of 104 cardiovascular prediction models, 91% carried a risk of harm at some plausible threshold (Gulati et al., 2022).

So the practice is right to check calibration. Our claim is narrower and, we think, more awkward: the standard check does not do what it is believed to do once the model has been penalized, and the direction of the error is systematic. It manufactures evidence of misfit out of the estimator rather than the model.

It may be objected that the Hosmer–Lemeshow test is no longer the right diagnostic anyway, and that a flexible calibration curve with its slope and intercept is to be preferred. That objection is well taken and it does not dissolve the problem. The displacement described here is a property of the estimator, not of the grouping: it is visible in the

calibration slope of the glaucoma fit, which is 2.43 — an apparent miscalibration produced entirely by the penalty and not by any defect of the model. We treat the grouped test because it is the diagnostic that carries a reference distribution, and therefore the one whose p -value can be wrong; a calibration curve inherits the same displacement but reports it as a picture rather than as a verdict. The same is true of the smoothing-based test of le Cessie and van Houwelingen (1991), which is a curve with a reference distribution attached and would inherit the displacement in both roles.

The paper is organized as follows. Section 2 shows why the reported p -value is not valid and separates two null hypotheses that coincide under maximum likelihood but not under penalization. Section 3 derives the corrected reference distribution and obtains a valid procedure. Section 4 gives the law governing what such a test can and cannot detect once the linear predictor must be estimated. Section 5 reports the simulation evidence, and Section 6 returns to the glaucoma model and states what the corrected analysis concludes.

2. Why the reported p -value is not valid

2.1. Notation

Let $y_i \in \{0, 1\}$ and $\mathbf{x}_i \in \mathbb{R}^p$, $i = 1, \dots, n$, and write X for the $n \times (p + 1)$ design matrix including the intercept column. The logistic model specifies $\pi_i(\boldsymbol{\beta}) = \{1 + \exp(-\mathbf{x}_i^\top \boldsymbol{\beta})\}^{-1}$. Penalized estimation solves

$$\hat{\boldsymbol{\beta}} = \arg \max_{\boldsymbol{\beta}} \left\{ \ell(\boldsymbol{\beta}) - \frac{1}{2} \lambda \boldsymbol{\beta}^\top D \boldsymbol{\beta} \right\}, \quad D = \text{diag}(0, 1, \dots, 1), \quad (1)$$

the intercept being left unpenalized. The scale of λ matters and is a common source of confusion: the widely used implementation of Friedman et al. (2010) minimizes $-n^{-1} \ell(\boldsymbol{\beta}) + \frac{1}{2} \lambda_g \|\boldsymbol{\beta}_{-0}\|^2$, so that $\lambda = n \lambda_g$. We verified this identity numerically to 1.6×10^{-9} and report both scales throughout.

Write $W = \text{diag}\{\hat{\pi}_i(1 - \hat{\pi}_i)\}$, $F = X^\top W X$, $K = \lambda D$ and $M = F + K$. Let C be the $G \times n$ matrix of indicators of the G equal-frequency groups of the fitted probabilities, $V = C W C^\top$, and

$$\mathbf{r} = V^{-1/2} C(\mathbf{y} - \hat{\boldsymbol{\pi}}), \quad U = V^{-1/2} C W X. \quad (2)$$

The Hosmer–Lemeshow statistic is $\|\mathbf{r}\|^2$ up to the choice of denominator.

Two bases for the grouped residuals appear throughout, and everything we prove applies to both. The *decile* statistic is $\|\mathbf{r}\|^2$ itself, which spends all G directions equally. The *EDGE* statistic of Ebrahim and El-Kotory (2026) instead projects the residual vector onto the orthogonal polynomials of degree one to three in the group-mean fitted probability: writing Z for the resulting $G \times 3$ matrix, it is

$$(Z^\top \mathbf{r})^\top (Z^\top Z)^{-1} (Z^\top \mathbf{r}). \quad (3)$$

It spends three directions rather than G , and it is aimed at smooth departures along the fitted index rather than at arbitrary ones. We report both because they behave differently in high dimension, and that difference is itself informative.

2.2. Two null hypotheses that separate under penalization

Two distinct questions are being asked, and ordinary practice does not distinguish them because ordinarily they cannot be distinguished. Write

$$H_0^{\text{str}} : \text{the logistic structure is correct, that is } \Pr(y_i = 1 \mid \mathbf{x}_i) = \pi_i(\boldsymbol{\beta}_0) \text{ for some } \boldsymbol{\beta}_0; \quad (4)$$

$$H_0^{\text{cal}} : \text{the deployed predictor is calibrated, that is } \mathbb{E}(y_i \mid \hat{\pi}_i) = \hat{\pi}_i. \quad (5)$$

Under maximum likelihood at fixed dimension, $\hat{\boldsymbol{\beta}} \rightarrow \boldsymbol{\beta}_0$, so (4) implies (5) asymptotically and the distinction is empty. The qualification matters and we return to it in Section 2.4(c): once p/n is bounded away from zero the maximum likelihood estimator is itself displaced, and the two hypotheses separate even without a penalty. Under

penalization it is not: $\hat{\beta}$ converges to a shrunken pseudo-parameter, so a model whose structure is exactly right delivers a predictor that is exactly *miscalibrated*, by an amount the analyst chose when selecting λ .

The second hypothesis is calibration in the sense of Dawid (1982), and sits at the “moderate” level of the hierarchy of Van Calster et al. (2016), which separates mean calibration and calibration in the large from the conditional statement in (5). This matters for reading the output. A test that rejects has told us that (5) fails; practitioners read it as evidence that (4) fails. Under penalization those are different statements, and the whole difficulty of this paper lies in that gap. Our object is a valid test of (4). A test of (5) is also useful — it is what one wants before deployment — but it is answered by recalibration, not by rebuilding the model.

2.3. The corrected law

Proposition 1 *Under the regularity conditions given in the supplementary material, with the penalty contributing $K \succeq 0$ to the curvature,*

$$\hat{\beta} - \beta = M^{-1}\{X^\top(\mathbf{y} - \boldsymbol{\pi}) - K\beta\} + o_p(1),$$

and consequently $\mathbf{r} \sim \mathcal{N}(\boldsymbol{\mu}_K, \Omega_K)$ with

$$\Omega_K = I_G - UM^{-1}U^\top - UM^{-1}KM^{-1}U^\top = I_G - UM^{-1}(F + 2K)M^{-1}U^\top, \quad (6)$$

$$\boldsymbol{\mu}_K = UM^{-1}K\beta + (\text{second-order remainder}). \quad (7)$$

Two things change at once. They push the test the same way, towards over-rejection, but they are of very different sizes.

The covariance moves away from its maximum likelihood value $I_G - UF^{-1}U^\top$, and it moves *upward*. Writing $M = F + K$ and using $F^{-1} - M^{-1} = M^{-1}KF^{-1}$ together with $M^{-1}(F + 2K)M^{-1} = M^{-1} + M^{-1}KM^{-1}$,

$$F^{-1} - M^{-1}(F + 2K)M^{-1} = M^{-1}K(F^{-1} - M^{-1}) = M^{-1}KF^{-1}KM^{-1} \succeq 0, \quad (8)$$

so $\Omega_K \succeq I_G - UF^{-1}U^\top$. The classical covariance *understates* the residual variance once a penalty is applied, which by itself would make a test referred to it anti-conservative. The effect is real but small: the supplementary material measures $\text{tr}(\Omega_K)$ exceeding $\text{tr}(I_G - UF^{-1}U^\top)$ by 0.23 in design A and by 0.92 in design B, out of ten groups. A direct decomposition confirms that it is not what breaks the test: correcting the covariance alone, leaving the non-centrality in place, moves the rejection rate from 0.994 to 0.992 at $\lambda = 137$ in design A. It also needs no separate remedy, because Proposition 2 shows that once the non-centrality is subtracted the covariance returns to $I_G - UF^{-1}U^\top$ exactly.

What is left, and what drives the distortion, is (7). We have $\mathbb{E}(\mathbf{r}) \neq \mathbf{0}$ whenever $K\beta \neq \mathbf{0}$, so the correct reference is a *non-central* weighted chi-squared distribution. Shrinkage displaces the grouped residuals systematically, in a direction fixed by the penalty, and the classical test reads that displacement as evidence against the model. The displacement grows with λ ; the classical critical value does not.

Remark 1 This is not the Chernoff–Lehmann problem (Chernoff and Lehmann, 1954; Moore and Spruill, 1975), in which estimation shifts the null distribution of a chi-squared statistic while leaving it centred. Here the null distribution acquires a non-zero mean. Moore and Spruill’s unified theory assumes a $\sqrt{n}$ -consistent estimator satisfying an orthogonality condition; a penalized estimator with λ growing satisfies neither, which is precisely why their conclusions do not transfer.

2.4. How large the error is

Three features of Figure 1 deserve comment.

(a) The distortion is not a small-sample curiosity; it reaches certainty. A test whose size is 1.000 carries no information at all: its p -value is a deterministic function of the penalty, not of the data.

(b) It occurs at the penalties cross-validation actually chooses. Over 500 replications the one-standard-error rule exceeded $\lambda_g = 0.05$ in every replication of both designs, and $\lambda_g = 0.05$ is already the column at which design B has reached 0.990.

The useful summary is a comparison of two numbers. Across the twelve configurations of Figure 1, the uncorrected test rejects a correct model more often than not once N exceeds d , and this rule is right in eleven of the twelve. Averaged over 200 designs per cell, the predicted sizes track the measured ones with a mean absolute error of 0.069.

Two features of (9) explain why dimension is so damaging. N grows like λ^2 for small λ and saturates at $\|U\beta\|^2$; and d falls as p/n rises, from 7.97 in design A to about 6.0 in design B. Dimension moves both terms the wrong way at once, which is the analytic reason design B is catastrophic where design A is merely bad.

We are explicit about what this screen does not do. It answers “do I need to worry?”, not “is my model wrong?”, and it does not replace the corrected test. Its approximation is first order, and in design A at heavy shrinkage it is optimistic — 0.76 against a measured 0.92. More importantly it cannot see the Sur and Candès (2019) offset at all: with no penalty $N = 0$, so it predicts exactly 0.05 where design B in fact rejects 17.8% of correct models. The screen therefore *understates* the problem in high dimension and never overstates it, which is the safe direction for a device whose purpose is to raise an alarm.

3. The shrinkage-corrected Hosmer–Lemeshow test

3.1. Removing the displacement

Proposition 1 suggests an obvious remedy: estimate the shrinkage non-centrality μ_K and subtract it. We call the resulting procedure the *shrinkage-corrected Hosmer–Lemeshow test*, and the remedy turns out to do more than it was asked to.

Proposition 2 Let $\tilde{\beta} = \hat{\beta} + F^{-1}K\hat{\beta}$ and $\hat{\mu} = UM^{-1}K\tilde{\beta}$. Then, to first order,

$$\text{var}(\mathbf{r} - \hat{\mu}) = I_G - UF^{-1}U^\top,$$

the ordinary maximum likelihood covariance.

We give this proof in the text, rather than in the supplement with the others, because it is short and because it is the reason the method needs no new reference distribution.

Proof Since $I + F^{-1}K = F^{-1}(F + K) = F^{-1}M$, the debiased estimate is $\tilde{\beta} = F^{-1}M\hat{\beta}$. Substituting the expansion of Proposition 1 and using $F^{-1}M\beta = \beta + F^{-1}K\beta$, the two penalty terms cancel and

$$\tilde{\beta} = \beta + F^{-1}X^\top(\mathbf{y} - \pi) + o_p(1), \quad (10)$$

which is the first-order expansion of the *unpenalized* maximum likelihood estimator in the regime of the regularity conditions. Where that estimate does not exist, as on the data of Section 1, $\tilde{\beta}$ should be read not as an estimator in its own right but as a one-step undoing of the shrinkage the penalty imposed. The same expansion writes the grouped residuals as $\mathbf{r} = A(\mathbf{y} - \pi) + UM^{-1}K\beta + o_p(1)$ with $A = V^{-1/2}C - UM^{-1}X^\top$. Subtracting $\hat{\mu} = UM^{-1}K\tilde{\beta}$ removes the non-random term exactly and leaves

$$\mathbf{r} - \hat{\mu} = \{V^{-1/2}C - UM^{-1}(I + KF^{-1})X^\top\}(\mathbf{y} - \pi) + o_p(1).$$

The cancellation is now algebraic:

$$M^{-1}(I + KF^{-1}) = M^{-1}(F + K)F^{-1} = M^{-1}MF^{-1} = F^{-1}. \quad (11)$$

Writing $A^* = V^{-1/2}C - UF^{-1}X^\top$ and using $\text{var}(\mathbf{y} - \pi) = W$ together with $V^{-1/2}CWC^\top V^{-1/2} = I_G$, $V^{-1/2}CWX = U$ and $X^\top WX = F$ gives $A^*WA^{*\top} = I_G - UF^{-1}U^\top$. $\square$

Two things are worth drawing out. The estimation noise contributed by the plug-in exactly cancels the penalization of the covariance, and (11) shows this is an identity rather than an approximation. So the practitioner does not have to learn a new reference distribution: once the displacement is removed, the familiar one is correct again. And (10) says what $\tilde{\beta}$ is — the one-step correction that undoes, to first order, exactly the shrinkage the penalty imposed. That is also the one place the procedure is fragile, since it inverts F .

Remark 2 *Nothing in either proposition uses the linearity of the ridge penalty. The proof needs the penalty only through its contribution to the estimating equation, so if a smooth penalty contributes a known deterministic $\mathbf{a}(\beta)$ with $K(\beta) = -\partial\mathbf{a}/\partial\beta$, then $\mu = UM^{-1}\mathbf{a}(\beta)$ and $\tilde{\beta} = \hat{\beta} + F^{-1}\mathbf{a}(\hat{\beta})$, and the cancellation (11) goes through unchanged, because it only ever used $M^{-1}(F + K)F^{-1} = F^{-1}$. Ridge is the case $\mathbf{a}(\beta) = K\beta$. This covers the Liu estimator, the ridge part of the elastic net, and any Bayesian maximum a posteriori fit with a smooth prior. It also covers, formally, Firth's correction (Firth, 1993), whose score contribution is the Jeffreys term — which matters here because Firth, not ridge, is the standard response to separation in clinical prediction (Puhr et al., 2017). We state that as a formal consequence and not as a verified one: Firth's $\mathbf{a}(\beta)$ is not linear and its $K(\beta)$ is not constant, so the same first-order remainder caveat applies with more force, and we have not run the numerical check that would let us recommend it.*

There is also a reading of $\tilde{\beta}$ that connects this construction to a literature it might otherwise seem separate from. The penalized stationarity condition is $X^\top(\mathbf{y} - \hat{\pi}) = K\hat{\beta}$. Substituting that into the generic one-step debiased estimator $\hat{\beta} + F^{-1}X^\top(\mathbf{y} - \hat{\pi})$ returns $\hat{\beta} + F^{-1}K\hat{\beta}$, which is $\tilde{\beta}$. So $\tilde{\beta}$ is the debiased estimator of Zhang and Zhang (2014) and van de Geer et al. (2014), specialised to ridge — and in the ridge case two things that are approximations in that literature become exact: the score residual is a deterministic function of $\hat{\beta}$, and F^{-1} is available directly rather than through a nodewise-lasso approximation. That is why the cancellation of Proposition 2 is an identity here and not an approximation.

It is worth saying how that fragility does and does not show up, because the obvious diagnostic is the wrong one. On the glaucoma data the cross-product matrix has condition number about 9×10^7 , and $\|\tilde{\beta}\|$ exceeds $\|\hat{\beta}\|$ by a factor between 618 and 1637 across the four penalties. Read as an estimate, $\tilde{\beta}$ is meaningless there. But $\tilde{\beta}$ is never read as an estimate: it enters only through the generator $\pi(\tilde{\beta})$, and the generator is well behaved. At the cross-validated penalty its fitted probabilities span $[0.0009, 0.997]$ with standard deviation 0.34, not one of the 196 observations lies within 10^{-6} of 0 or 1, and it implies 96.7 events against the 98 observed. The large components of $\tilde{\beta}$ lie in directions in which the design has almost no variance, so they inflate its norm and leave $X\tilde{\beta}$ almost unchanged. The quantity a practitioner should check before trusting this procedure is therefore the spread of the generator, which is computable on any dataset, and not the norm of $\tilde{\beta}$.

3.2. Prepivoting

The first-order correction is not enough at the penalties cross-validation selects. It removes the anti-conservative failure but leaves the test conservative, because $\hat{\mu}$ is itself estimated at a penalty large enough that the first-order account of its error is inadequate.

We therefore refer the corrected statistic to a reference built from itself. The procedure is:

- (a) fit the penalized model at λ and compute $S = \|\mathbf{r} - \hat{\mu}\|^2$;
- (b) form the generator $\pi(\tilde{\beta})$ from the debiased estimate;
- (c) for $b = 1, \dots, B$, draw $\mathbf{y}_b^*$ from that generator, refit the penalized model at the same λ , and recompute the same corrected statistic S_b^* ;
- (d) report $\{1 + \#(S_b^* \geq S)\}/(B + 1)$.

Because the statistic is corrected in both the real and the bootstrap world, the reference inherits the same finite-sample imperfections as the observed value and cancels them. This is the prepivoting principle of Beran (1987, 1988): one does not need the statistic to be asymptotically pivotal, only to be referred to its own sampling distribution under a generator that is correct to the order that matters.

Three parts of this procedure carry the argument, and two of them have a near neighbour that fails. The generator must be $\pi(\tilde{\beta})$: generating from $\pi(\hat{\beta})$ builds the shrinkage into the null world, and the test becomes conservative. The statistic must be corrected in both worlds, since an uncorrected bootstrap statistic and a corrected observed one are not the same quantity. And we hold λ fixed in the bootstrap world, so that the null world's penalty is not itself random. The catalogue of neighbouring constructions, with the size of each failure, is in the supplementary material. It is also worth stating that an oracle which knows β_0 cannot be built on a bootstrap reference at all, because the truth of the bootstrap world is $\tilde{\beta}$ and not β_0 ; oracle comparisons in Section 4 therefore use known-null Monte Carlo instead.

4. What the test can and cannot detect

A valid test is not automatically a useful one. This section gives the law that governs its power, and we present it as a contribution rather than as a limitation, because it applies to every test in this family and not only to ours.

4.1. The visible effect size

Grouping is performed on the *fitted* linear predictor $\hat{\eta}$, so the only part of a departure that a grouped test can see is the part that survives conditioning on $\hat{\eta}$. For a departure $\gamma g(\mathbf{x})$ added to the linear predictor, define the visible effect size

$$\tau = \gamma \frac{\text{sd}[\mathbb{E}\{g(\mathbf{x}) \mid \hat{\eta}\}]}{\text{sd}(\hat{\eta})}. \quad (12)$$

The nominal coefficient γ is not comparable across designs — the same γ means different things when $p = 5$ and when $p = 100$ — whereas τ is. When power is plotted against $\nu\sqrt{n}$, where $\nu = \tau \text{sd}(\hat{\eta})$, curves from designs of very different dimension collapse onto one another (Figure 3, panel A; mean absolute gap 0.017 against a Monte Carlo standard error of 0.023 for a single difference).

There is a subtlety worth making explicit, because it corrects an expectation one naturally forms from Proposition 3. One might expect a residual dimensional offset to remain visible on a τ axis. It does not, and should not: the numerator of (12) already contains the factor ρ^k , so τ absorbs the attenuation by construction. The law of Section 4.2 is untouched; what it governs is how much τ a given γ buys, not the shape of the curve once τ is on the axis.

There is also a limit beyond which *any* grouped test may lose power as the departure grows, and it is worth stating because we have not seen it stated. A cubic departure in the index standardizes to $Z + \text{rel} \cdot (Z^3 - 3Z)/\sqrt{6}$ with $\text{rel} = \gamma/\text{sd}(\eta_0)$, whose derivative at $Z = 0$ is $1 - 3\text{rel}/\sqrt{6}$. Once $\text{rel} \geq \sqrt{6}/3 = 0.82$ the true event probability is no longer monotone in the linear predictor, so sorting patients by predicted risk no longer sorts them by the departure, and the power of any test that groups on the index can fall as the departure grows. Of the cells in Figure 3, one point of the design A cubic series lies beyond this threshold ($\text{rel} = 0.97$) and three of the design B cubic series do (1.05, 1.57, 2.19). Power nevertheless rises monotonically across the plotted range, so the threshold bounds the interpretation rather than invalidating the curves. This is a property of grouping on a monotone summary, shared by the Hosmer–Lemeshow family and by every grouped test; we give its threshold because it is easy to cross without noticing.

4.2. The attenuation law

Proposition 3 Write η^* and $\hat{\eta}^*$ for the true and the fitted linear predictor, each standardized to mean zero and unit variance, and suppose the pair is jointly Gaussian with correlation ρ . Since $\mathbb{E}\{\text{He}_k(\eta^*) \mid \hat{\eta}^*\} = \rho^k \text{He}_k(\hat{\eta}^*)$ for the Hermite polynomials He_k , a departure equal to the degree- k component of the true linear predictor survives grouping by the fitted predictor with amplitude ρ^k , and its non-centrality is attenuated by ρ^{2k} .

The consequence is sharp and, once stated, unsurprising. Estimation error in the index costs a linear departure a factor ρ^2 , a quadratic departure ρ^4 , and a cubic departure ρ^6 . At $\rho = 0.9$ — a fitted index that most analysts would regard as excellent — a cubic departure retains only 53% of its non-centrality; at $\rho = 0.7$, only 12%. Higher-order departures are therefore intrinsically hard to see through an estimated index, for *any* test that groups on that index.

4.3. Two mechanisms, not one

It is tempting to collapse the high-dimensional difficulty into a single statement about p/n . That would be wrong, and the distinction is practically useful because the two mechanisms have different remedies.

(a) *Dilution* is a property of the design, not of the dimension. A departure attached to one coordinate, such as $\gamma(x_1^2 - 1)$, contributes to the index in proportion to that coordinate's share of the index variance. In our design B that share is small, and the visible *variance* of the departure is accordingly smaller by a factor of 16 than in design A. Dilution would occur at the same magnitude in a low-dimensional design in which one coordinate happened to carry a small weight.

(b) *Index-estimation noise* is the dimensional mechanism, and it is Proposition 3: as p/n grows, ρ falls, and every departure is attenuated by ρ^{2k} .

The two are separable by experiment, and the experiment separates them cleanly. Group the residuals by the *true* index instead of the fitted one, holding everything else fixed, and calibrate each arm by direct Monte Carlo from the true model so that both are exactly correct in size by construction. Whatever power then changes is attributable to the grouping alone. In design B this gives Table 1.

The contrast is the whole argument of this section. For the index-aligned departure, knowing the true index raises power from 0.092 to 0.623 on the decile basis and from 0.046 to 0.633 on the EDGE basis, a factor of nearly fourteen. For the coordinate-wise departure it changes nothing: 0.059 to 0.067, and on the EDGE basis 0.061 to 0.059. So mechanism (b) is real and large, but it is not what limits the coordinate-wise case; there the departure is diluted before any index is estimated, and an oracle who knew the index exactly would be no better off. Conversely an index-aligned departure suffers no dilution at all, and its power curves collapse across designs once expressed in $\nu\sqrt{n}$.

The honest scope statement is therefore not that the test lacks power in high dimensions. It is that a grouped test on an estimated index detects index-aligned departures at a rate governed by $\nu\sqrt{n}$ in any dimension, and detects coordinate-wise departures at a rate that also depends on how much of the index that coordinate carries.

4.4. A division of labour, measured

Tests for high-dimensional generalized linear models built by other routes (Guo and Chen, 2016; Shah and Bühlmann, 2018; Janková et al., 2020; Javanmard and Mehrabi, 2024) are usually described as alternatives to a test like ours. We ran the comparison rather than assert it. In design B at $\lambda = 416$ with $G = 10$, fully paired so that every method sees the same dataset in every replicate, the residual-prediction test of Guo and Chen (2016) beats ours decisively on the coordinate-wise quadratic departure — 0.710 at $\gamma = 1.0$ and 1.000 at $\gamma = 1.5$, against our 0.055 to 0.090 — and the two are level on the cubic.

That result is worth reporting for its own sake, but it is not the interesting one. The two tests are not noisy versions of each other: *they reject on different datasets*. Across all six configurations the number of replicates in which both tests reject is between 0 and 15 out of 200, and the paired differences are -0.655 and -0.925 with $p < 2 \times 10^{-16}$. A residual-prediction method asks whether the model is right as a function of x ; ours asks whether the deployed predictor is calibrated along its own index. Those are different questions, and the data say so.

Our loss on the coordinate-wise departure is therefore confirmation of this paper's own dilution mechanism, arriving from outside it: x_1 carries 3.4% of the variance of the index at $p = 100$, so a departure attached to x_1 is nearly invisible to any test that groups on the index, and plainly visible to one that does not. An analyst who suspects a specific covariate should use a residual-prediction test. One who wants to know whether the risks the model reports can be read as risks should use this one.

Three points of software provenance, because they are checkable. The Shah–Bühlmann package is `RPtests`, and it targets the Gaussian linear model; on a binary response it is off-label, and we ran it only so that its size could speak. The Guo and Chen (2016) implementation `GRPtests` was removed from the Comprehensive R Archive Network and archived on 8 May 2022; we installed it from the archived tarball, and we ran it both at its package default and at a single split, so that it is not handicapped by a tuning choice. GRASP has no package and was not attempted.

5. Simulation

Two designs are used throughout. Design A has $n = 500$, $p = 5$ and independent standard normal covariates; design B has $n = 400$, $p = 100$ and an autoregressive covariate correlation of 0.7, giving $p/n = 0.25$. In both, the null model is logistic with the stated coefficients, so every rejection under the null is a false one. Full specifications, seeds and code are in the supplementary material.

5.1. Size

Table 2 is the central empirical claim of the paper. Against an uncorrected test that rejects correct models between 92% and 100% of the time on these same six configurations, the prepivoted corrected test holds its nominal level. In design A it does so at every penalty on both bases, with standard errors of 0.005.

One cell requires comment rather than concealment. On the EDGE basis in design B at the lightest penalty, $\lambda = 50$, the test is detectably conservative: 0.0370 with an exact 95% interval of [0.0314, 0.0433], which is 4.4 standard errors below nominal. We report this cell at four times the replication of the others because our first estimate of it, from a single block of 1000, was 0.0270, which would have placed it outside the range [0.03, 0.08] we fixed in advance as acceptable. The pooled estimate lies inside that range; it is the superseded single block that did not. Two further independent blocks did not reproduce that value, the three are homogeneous ($p = 0.09$), and the pooled estimate is the one given. The episode is a caution about reading a single block at a probability this small, and we apply the same standard to any other cell that falls outside the range.

The conservatism that survives is genuine but modest, and it is not uniform — the same basis returns 0.0540 at $\lambda = 416$ and 0.0610 at $\lambda = 1000$ — so it is confined to the light-penalty end of the high-dimensional design and should not be described as a general property of the basis. It does mean that where the EDGE basis is compared with the decile basis in design B, a size-adjusted comparison is the fair one, and we report both.

5.2. Power

The comparison that matters is not corrected-versus-uncorrected under the null, where the uncorrected test is disqualified, but corrected-versus-classical where the classical test is legitimate. We therefore compare the corrected test on a penalized fit with the classical test on a maximum likelihood fit in design A, where the latter exists. Paired over the eight design A alternatives, the correction costs what Figure 2 and Table 4 report, and it is *not* free. On the decile basis it costs about two and a half percentage points, or thirteen per cent of the classical test's power, and at 2000 replications that is unambiguous. We record this plainly because an earlier version of this experiment, at 200 replications, returned -0.010 with $p = 0.10$ and would have supported the more comfortable claim that the cost is indistinguishable from zero. It was simply too small to see the effect.

Two things put the cost in proportion. It is a cost paid only where there is a choice. In design B no valid maximum likelihood comparator is available, because the unpenalized fit is itself invalid there; and on the glaucoma data the maximum likelihood estimate does not exist at all. In both the classical test is unavailable at any price, and the comparison has no second column. And it is much smaller than the gain available from the basis. Switching from the decile to the EDGE basis under the same correction raises mean power across these eight alternatives from 0.167 to 0.281, an increase of 68% — roughly five times what the correction costs. An analyst who wants the power back does not need a different test; they need a better basis.

Design B tells a blunter story. Over the range $\gamma \leq 1.5$ the corrected test has no detectable power at all: the decile rejection rate moves from 0.0640 to 0.0740 between $\gamma = 0$ and $\gamma = 1.5$, a change of 0.010 against a standard error of 0.0075. This too corrects the smaller study, which showed 0.040 rising to 0.095 and looked like the beginning of a power curve. It was noise. The departure has to be made several times larger before design B responds at all, for the dilution reason given in Section 4.3.

Figure 3 shows what governs power once dimension enters. It is built on a different calibration from Tables 2–4, and we say so because the difference matters for reading it. Its arms are calibrated by direct Monte Carlo from the true model rather than by prepivoting — 2000 null draws per cell, with the critical value taken at the 95th percentile — so every curve starts from exactly 0.05 by construction. That is the right design for the question the figure asks — which departures a grouped test can see, and on what scale — because it removes the calibration of the test as a competing explanation for any difference between curves. It does mean the figure is not a display of the prepivoted procedure itself; for that, see Tables 2 and 3.

6. The glaucoma model, reconsidered

We now return to the model of Section 1 and apply the corrected test.

6.1. What the uncorrected test was reacting to

Figure 4 shows the calibration of the ridge fit at the cross-validated penalty. The pattern is not random. In the lowest deciles of predicted risk the observed event rate lies *below* the diagonal — risk is predicted too high — and in the highest deciles it lies *above* it, where risk is predicted too low. The fitted probabilities are therefore pulled systematically towards the centre, in nine of the ten deciles and in the direction the penalty dictates; the sixth sits

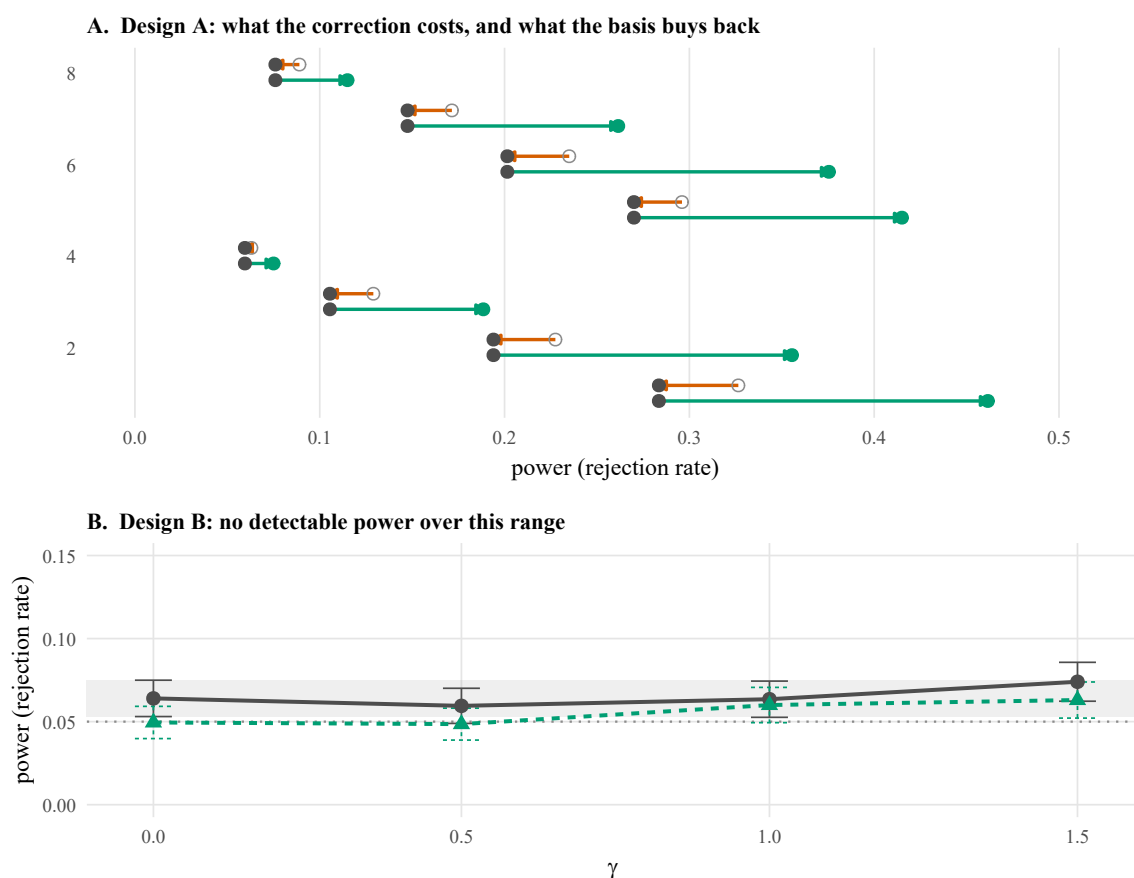


Figure 2 The correction costs power, and the choice of basis buys back more than it costs. In panel A each alternative occupies two half-rows: above, an orange arrow runs from the classical test on a maximum likelihood fit (open circle) back to the shrinkage-corrected test on the decile basis (filled grey circle), and its length is the cost; below, a green arrow runs from that same point forward to the EDGE basis (filled green circle), and its length is the gain. The two are drawn with identical weight so that the lengths compare directly. Averaged over the eight alternatives the cost is 0.0250 and the gain 0.1138, about four and a half times larger. Panel B is design B on a deliberately tight vertical scale, with two-standard-error bars and a grey band at the $\gamma = 0$ rejection rate plus or minus two standard errors; every alternative lies inside that band, so there is no detectable power over this range. Based on 2000 replications per cell.

on the diagonal, at the crossing point. The apparent calibration slope is 2.43, meaning the linear predictor would have to be multiplied by roughly two and a half before the fitted probabilities agreed with the observed rates.

That compression is not a defect of the model. It is what a penalty does, and it is the price paid for obtaining any estimate at all in a design where the maximum likelihood estimate does not exist. It is the displacement that Proposition 1 describes. The uncorrected test sees it, has no way to distinguish it from misspecification, and reports the model as wrong. We say it is the displacement Proposition 1 describes rather than claim that the proposition applies here: its conditions hold the dimension fixed, and these data have $p/n = 0.32$. What the proposition supplies on this dataset is the mechanism and its direction; what supplies the p -values is the pre-pivoted procedure, whose validity in this regime rests on the simulation evidence of Section 5.

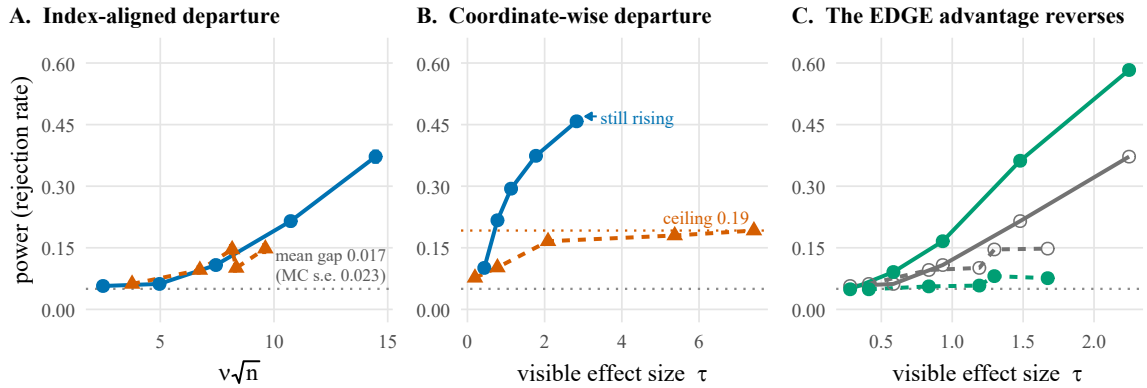


Figure 3 Power is governed by the visible effect size, not the nominal one: on the decile basis, curves from designs of very different dimension collapse when the departure is aligned with the index (A), but a coordinate-wise departure is diluted and its power flattens in high dimension (B), and the advantage of the EDGE basis reverses with dimension (C). In panels A and B the solid blue curve with circles is design A and the dashed orange curve with triangles is design B; in panel C the green curve with filled circles is the EDGE basis and the grey curve with open circles the decile basis, solid for design A and dashed for design B. Rejection rates in all three panels use a critical value obtained by direct Monte Carlo from the true model rather than by pre pivoting, so that every cell is exactly 5% under the null by construction and the comparison isolates the grouping; $B_1 = 1000$ replications per cell, Monte Carlo standard error at most 0.016, shown as error bars in panel A. This is a separate experiment from Tables 2 and 3, which use a pre pivoted bootstrap reference. In panel A the third and fourth design B points sit at almost the same value of $v\sqrt{n}$ (8.16 and 8.34), because v is not monotone in γ ; the apparent dip between them is that, not a loss of power. One point of the design A cubic series and three of the design B cubic series lie beyond the monotonicity threshold of Section 4.1.

6.2. The corrected analysis

Table 5 reports both tests across four penalties and three choices of the number of groups. Three things should be said about it, in order of how much they matter.

(a) The uncorrected test is uninformative here. It rejects in every one of the twelve configurations, with p below 0.006 throughout and below 10^{-5} at and beyond the cross-validated penalty. Given the size distortion documented in Section 5 — in simulation designs of comparable p/n at comparable shrinkage, rejection of correctly specified models between 92% and 100% of the time (Table 2) — a small p -value from this test carries almost no information about this model.

We use the sharpest available form of the uncorrected test, referring the grouped statistic to the maximum likelihood covariance rather than to χ^2_{G-2} . The distinction matters more than one would expect, and it illustrates the paper's own point from a different angle. The classical degrees-of-freedom count would be $G - 2 = 8$ at ten groups, so it allows two decile directions to be absorbed by estimation. The trace of $I_G - UF^{-1}U^\top$ on these data is 4.84 at the cross-validated penalty, and lies between 4.84 and 5.08 across the four penalties, so more than five of the ten directions are absorbed. Fitting 62 penalized coefficients costs far more of the residual space than the classical count admits. The textbook χ^2_{G-2} reference is therefore too diffuse, and gives p -values from 7.5×10^{-7} to 0.115 across the same grid, failing to reject in one cell of the twelve. Comparing our correction against that weaker version would flatter it, so we do not.

(b) The corrected test returns $p \approx 0.03$ at the cross-validated penalty with ten groups, and values between 0.014 and 0.678 across the grid. The evidence for misfit is therefore real but modest, and it is weaker than the uncorrected analysis suggested by more than three orders of magnitude. We deliberately do not claim that the corrected test exonerates the model. It does not. What it does is replace an uninterpretable number with an interpretable one.

(c) The corrected p -value varies with the number of groups, from 0.026 at ten groups to 0.172 at twenty. Sensitivity to grouping is inherited from the Hosmer–Lemeshow family and our correction does not remove it; Nattino et al. (2020) address it directly by replacing the grouping with a fixed partition of the probability scale,

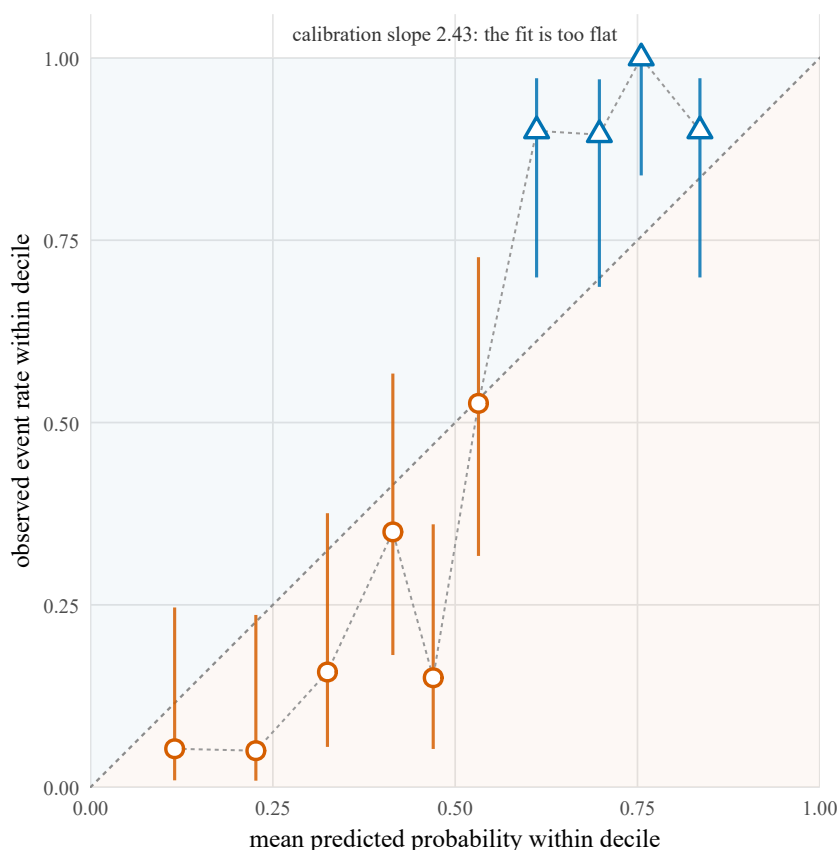


Figure 4 The ridge fit is not mis-ordered, it is too flat: shrinkage pulls the predicted risks towards the centre, and it is that displacement, not misspecification, that the uncorrected test is reading. Deciles below the diagonal, drawn as orange circles, have their risk predicted too high; those above it, drawn as blue triangles, have it predicted too low. Vertical bars are Wilson intervals and the dashed line is equality.

which is complementary to the correction developed here. We report the whole grid rather than a single cell, and we would encourage the same practice generally.

6.3. A second dataset, and a different verdict

A reader is entitled to ask whether one dataset carries this much weight. We therefore ran the identical grid on *GlaucomaMVF* (package *ipred*), a second series of $n = 170$ eyes described by $p = 63$ of the same morphometric variables, measured with the same instrument, so that $p/n = 0.371$. It is not an independent replication — it is the same clinical question recorded with the same protocol — but it is a different sample, and the maximum likelihood estimate does not exist there either.

The outcome is not the same, and that is the point. On *GlaucomaM* the correction turns an overwhelming verdict into a borderline one and does not exonerate the model. On *GlaucomaMVF* it produces a clean reversal: the uncorrected test rejects in all twelve configurations on the EDGE basis, and the corrected test rejects in *none* of them on either basis, returning p between 0.224 and 0.998 on the decile basis and between 0.298 and 0.874 on the EDGE basis. Repeating the cross-validated cell under five seeds leaves those conclusions unchanged. The full grid is given in the supplementary material.

Two different verdicts from one mechanism is what we would expect and what we want. The correction does not manufacture acceptance; it removes a displacement whose size depends on the penalty and the design, and what is

left over is whatever misfit the data actually contain. On one dataset that residue is real but modest; on the other there is none to find. An analyst who applied the uncorrected test to both would have concluded that both models were badly misspecified, and would have been wrong about at least one of them.

6.4. What changes for the clinician

The two analyses support different actions.

Under the uncorrected analysis the model is rejected outright. The natural responses are to abandon it, to rebuild it with different predictors, or to collect more data — and in a screening application where the alternative is subjective expert assessment of the same optic disc images (Coan et al., 2023), discarding a model with an area under the curve of 0.905 has a cost measured in missed early disease.

Under the corrected analysis the model is not rejected outright, and the residual evidence of misfit has a recognizable shape: the predicted probabilities are too flat, not mis-ordered. Discrimination is unaffected by that, so the model remains usable for ranking patients by risk — which is what a screening triage actually requires. Where the model should not be used unaltered is where the predicted probability is read as a number, for instance when it is compared against a fixed threshold to decide referral. The remedy there is recalibration rather than reconstruction, and the direction of the required correction is known from the fit itself.

This distinction has a reassuring consequence that is worth stating explicitly, because it is not obvious. Penalization compresses risks towards the average, which in the terminology of Van Calster and Vickers (2015) is underfitting rather than overfitting. Of the four kinds of miscalibration they examine, underfitting is the only one that never made a model clinically harmful in the sense of being worse than a default strategy. A penalized model that has been wrongly condemned by an uncorrected goodness-of-fit test is therefore likely to be one whose miscalibration was, in decision-analytic terms, the benign kind. The cost of the invalid test is not that it protects patients too zealously; it is that it discards usable models for the wrong reason.

6.5. A limitation of this application

These are development data, and every quantity we report is apparent rather than validated. Our purpose is a like-for-like comparison of two tests on the same fit, which does not require external validation; but the calibration slope of 2.43 should not be read as an estimate of what the model would do in a new population, and a model intended for clinical use would require external validation and, on this evidence, recalibration before deployment (Van Calster et al., 2020).

7. Software

The corrected test is implemented as `gof.pen()` in the R package `ebrahim.gof`, available from the Comprehensive R Archive Network. The EDGE basis of Ebrahim and El-Kotory (2026) is available through the same interface, and the EDGE statistic computed by the code used here agrees exactly with the package on three datasets. The simulations reported above use a self-contained implementation of the penalized fit, so that the replication archive does not depend on a package version; the scale identity $\lambda = n\lambda_g$ against Friedman et al. (2010) was checked numerically to 1.6×10^{-9} .

8. Discussion

The general lesson is the separation of Section 2.2. Whenever a fitting procedure deliberately biases the estimator — ridge, lasso, elastic net, Firth's correction, any Bayesian prior with a non-negligible influence — the structural null and the calibration null stop being the same hypothesis, and a diagnostic built for the first will read the deliberate bias as evidence against the model. Grouped goodness-of-fit tests are the case treated here because they are the ones clinical prediction papers actually use, but the mechanism is not specific to them: any diagnostic whose reference distribution is centred by an unbiasedness argument inherits the same problem.

Three things are deferred and we name them as open rather than solved.

(a) Our theory is first-order, treats λ as non-random and requires $\lambda = o(n)$, and holds the dimension fixed. The empirical behaviour we rely on most — validity at $p/n = 0.25$, and the glaucoma analysis itself at $p/n = 0.32$ —

comes from pre pivoting, for which we have no analytic account in the proportional regime where $p/n \rightarrow \kappa \in (0, 1)$. The results of Sur and Candès (2019) and Candès and Sur (2020) suggest that such an account exists and would be worth having.

(b) We treat the ridge penalty, and Remark 2 extends the argument to any smooth penalty. The lasso is genuinely outside it, but the reason is sharper than non-differentiability. For the lasso the stationarity condition reads $X^\top(\mathbf{y} - \hat{\boldsymbol{\pi}}) = \lambda \hat{\mathbf{s}}$ for a subgradient $\hat{\mathbf{s}}$, so the same substitution returns the debiased lasso rather than nothing at all. The obstruction is that $\hat{\mathbf{s}}$ is not a deterministic function of the fit on the active set, and F^{-1} must be replaced by a nodewise approximation; the displacement $\boldsymbol{\mu}$ therefore becomes random, and the exact cancellation of Proposition 2 no longer holds identically. Whether it holds to sufficient order is open, and we do not claim it.

(c) We fix λ throughout. In practice λ is chosen by cross-validation on the same data, and the resulting selection effect is not accounted for by our procedure or, so far as we know, by any other.

We are also explicit about what the test is. It is a test of calibration along the fitted index. It is not an omnibus test of the logistic structure, and Section 4 quantifies exactly how much of a departure it can be expected to see. We regard that quantification as part of the contribution: a diagnostic whose scope is known is more useful than one whose scope is assumed.

The last word belongs to the eyes. A model built to detect glaucoma early, from measurements a machine already takes, was reported by the standard check as badly misspecified — at every penalty, at every grouping, without exception. On that reading it would have been rebuilt or abandoned, and in a disease whose damage cannot be reversed, the cost of abandoning a usable screening model is counted in sight that was not saved. The corrected analysis says something different and more useful: the fitted probabilities are too flat, not mis-ordered; the ranking on which a screening triage depends is sound; and what the model needs before anyone reads its probabilities as numbers is recalibration, not reconstruction. That is a smaller and more actionable conclusion than the one the uncorrected test appeared to license, and it is the one the data actually support. We think it is also the more common situation. Wherever a prediction model has been penalized because it had to be, and then judged by a test that assumed it had not been, the verdict on record may be a verdict about the estimator rather than about the model.

Data availability

The GlaucomaM data are distributed in the R package TH.data on the Comprehensive R Archive Network. The predictors are standardized to zero mean and unit variance before fitting; the data are otherwise unmodified. The second dataset, GlaucomaMVF, is distributed in the R package ipred. All code, seeds and intermediate results required to reproduce every number and figure in this paper, together with the per-replication p -values described in the supplementary material, are available at <https://github.com/ebrahimkhaled/gof-penalized-paper> and archived at <https://doi.org/10.5281/zenodo.21900115>.

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Conflict of interest

None declared.

Table 1. Grouping on the fitted index against grouping on the true one, design B, with each arm calibrated to exact size by direct Monte Carlo so that the comparison is of power only. Knowing the index transforms the index-aligned departure and does nothing at all for the coordinate-wise one, which is the two mechanisms separated.

Departure	γ	Fitted index		True index	
		decile	EDGE	decile	EDGE
cubic (index-aligned)	1.5	0.092	0.046	0.623	0.633
quadratic (coordinate-wise)	1.0	0.059	0.061	0.067	0.059

Table 2. Rejection rate of the *prepivot corrected* test under a correctly specified model, at nominal level 0.05, with 149 bootstrap replicates. Design A uses 2000 Monte Carlo replications and design B uses 1000, except the $\lambda = 50$ row, which is pooled over three independent replication blocks totalling 4000 and is marked $\dagger$. Penalties are on the theory scale λ ; the corresponding λ_g is λ/n . The uncorrected test on these same six cells rejects between 0.92 and 1.00. With $B = 149$ the bootstrap p -value takes the values $k/150$, so 0.05 is not itself attainable; the adjacent attainable levels are $7/150 = 0.0467$ and $8/150 = 0.0533$.

Design	λ	Decile basis		EDGE basis	
		rate	s.e.	rate	s.e.
A	100	0.0435	0.0046	0.0500	0.0049
A	137	0.0475	0.0048	0.0505	0.0049
A	200	0.0465	0.0047	0.0505	0.0049
B	50 †	0.0595	0.0037	0.0370	0.0030
B	416	0.0710	0.0081	0.0540	0.0071
B	1000	0.0520	0.0070	0.0610	0.0076

Table 3. Power of the pre pivoted corrected test on the decile and EDGE bases, paired on identical fits and refits with the classical test on the maximum likelihood fit. Two departures are used: an omitted quadratic $\gamma(x_1^2 - 1)$ and an omitted interaction $\gamma x_1 x_2$. Based on 2000 replications (Monte Carlo standard error at most 0.0112). The $\gamma = 0$ row is size, and is the same simulation for both departures. No valid maximum likelihood comparator is available in design B. Size-adjusted power is given in the supplementary material; it differs from the entries below by at most 0.02 and reverses no comparison. Design A is at $\lambda = 100$ and design B at $\lambda = 416$. The $\gamma = 0$ rows come from a different run than the corrected-size table (T25_power_highB.R, stream 20260810, against T23_rerun.R, stream 20260807); the two agree to less than one Monte Carlo standard error.

Design	Departure	γ	Corrected test		Classical, on the MLE	
			decile	EDGE	decile	EDGE
A	—	0.00	0.0445	0.0460	0.0440	0.0540
A	quadratic	0.25	0.0760	0.1150	0.0890	0.1205
A	quadratic	0.50	0.1475	0.2615	0.1715	0.2675
A	quadratic	0.75	0.2015	0.3755	0.2350	0.3835
A	quadratic	1.00	0.2700	0.4150	0.2960	0.4080
A	interaction	0.25	0.0595	0.0750	0.0630	0.0870
A	interaction	0.50	0.1055	0.1885	0.1290	0.2065
A	interaction	0.75	0.1940	0.3555	0.2275	0.3780
A	interaction	1.00	0.2835	0.4615	0.3265	0.4885
B	—	0.00	0.0640	0.0495	—	—
B	quadratic	0.50	0.0595	0.0485	—	—
B	quadratic	1.00	0.0635	0.0600	—	—
B	quadratic	1.50	0.0740	0.0630	—	—

Table 4. What the correction costs in power. Each entry is the mean difference between the corrected test on a penalized fit and the classical test on a maximum likelihood fit, paired over the eight design A alternatives of Table 3 and computed from 2000 replications per cell. The cost is real on both bases and larger on the decile basis.

Basis	mean difference		t (7 d.f.)	relative
decile	−0.0250	(s.e. 0.0044)	−5.70, $p = 0.0007$	−13.0%
EDGE	−0.0115	(s.e. 0.0038)	−3.00, $p = 0.020$	−3.9%

Table 5. Uncorrected and corrected p -values for the glaucoma model, over four penalties and three group counts. The uncorrected test rejects in all twelve configurations; the corrected test does not. Penalties are on the theory scale, with $\hat{\lambda}_{1se} = 295.3$ and $\hat{\lambda}_{min} = 19.9$ selected by ten-fold cross-validation. The uncorrected column refers the grouped statistic to the maximum likelihood covariance $I_G - UF^{-1}U^T$ by Monte Carlo, which is the strongest form of the incumbent; entries below 10^{-5} are the resolution limit of 2×10^5 draws. Corrected p -values use 499 bootstrap replicates.

λ	Uncorrected			Corrected, decile basis			Corrected, EDGE basis		
	$G = 5$	$G = 10$	$G = 20$	$G = 5$	$G = 10$	$G = 20$	$G = 5$	$G = 10$	$G = 20$
$\hat{\lambda}_{min}$	0.0003	0.0005	0.0053	0.014	0.048	0.140	0.456	0.096	0.114
$\frac{1}{2}\hat{\lambda}_{1se}$	$< 10^{-5}$	$< 10^{-5}$	0.0001	0.032	0.070	0.062	0.094	0.032	0.106
$\hat{\lambda}_{1se}$	$< 10^{-5}$	$< 10^{-5}$	$< 10^{-5}$	0.036	0.026	0.172	0.218	0.034	0.078
$2\hat{\lambda}_{1se}$	$< 10^{-5}$	$< 10^{-5}$	$< 10^{-5}$	0.294	0.276	0.678	0.500	0.038	0.096

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